Supervyse OPF Supplement Pack Quick Start Guide

Revision 1.1

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Revision History

Revision Number	Date	Author	Reviewer	Summary
1.0	2025-02-11	Ben Chang	Conrad Yao	Initial version
1.1	2024-02-14	Ben Chang	Conrad Yao	Add a command to copy the aspeed-app file.

Introduction

This page introduces how to setup a build environment for compiling Supervyse OPF Supplement pack , including the software packages you need to install.

Steps to Install Insyde Supplement Pack

1.1.1 Download OpenBMC Foundation Codebase

- The Insyde Supplement Pack is built based on OpenBMC (commit id: **fad58ae33ca742da193e7c269f372eed5b63f11f**).
- This version of OpenBMC is sourced from GitHub:

  GitHub - openbmc/openbmc at fad58ae33ca742da193e7c269f372eed5b63f11f

1. Create and enter the SDK folder.

```
mkdir SDK
cd SDK
```

2. Download OpenBMC Foundation Codebase

```
git clone https://github.com/openbmc/openbmc.git
```

3. Enter the openbmc folder

```
cd  openbmc
```

4. Switch to the specified commit id

```
git checkout fad58ae33ca742da193e7c269f372eed5b63f11f
```

1.1.2 Copy the Insyde Supplement Pack to the OpenBMC Foundation Codebase

1. Extract OPF_RV24.12-35.25.03.9201_SupplementPack.tar

```
tar -xvf OPF_RV24.12-35.25.03.9201_SupplementPack_2025_02_14.tar
```

2. Copy **meta-insyde** to the OpenBMC Foundation Codebase.

```
cp -r OPF_RV24.12-35.25.03.9201_SupplementPack_2025_02_14/meta-insyde/
SDK/openbmc/
```

3. Copy **isc-src** to the OpenBMC Foundation Codebase.

```
cp -r OPF_RV24.12-35.25.03.9201_SupplementPack_2025_02_14/isc-src/
SDK/openbmc/
```

4. Copy **aspeed-app** to the OpenBMC Foundation Codebase.

```
cp -r OPF_RV24.12-35.25.03.9201_SupplementPack_2025_02_14/aspeed-app/
SDK/openbmc/meta-aspeed/recipes-aspeed/
```

Setup of the Build Environment

Before using the SDK, you should consider what your needs/requirements will be.
Supported Operating System list:

- Ubuntu 18.04/64 (EOS in June 2023)
- Ubuntu 20.04/64
- Ubuntu 22.04/64

Install the packages on your Ubuntu 18.04/64 & 20.04/64

Please install the following tools.

```
$ sudo apt-get update
$ sudo apt-get install -y git build-essential libsdl1.2-dev texinfo gawk chrpath diffstat zstd pigz
$ sudo apt-get install -y unzip gcc-multilib g++-multilib xxd curl liblz4-tool
$ sudo apt-get install -y python jq python3-pip
$ sudo -H pip3 install yq
```

Install the packages on your Ubuntu 22.04/64

Please install the following tools.

```
$ sudo apt-get update
$ sudo apt-get install -y git build-essential libsdl1.2-dev texinfo gawk chrpath diffstat zstd pigz
$ sudo apt-get install -y unzip gcc-multilib g++-multilib xxd curl liblz4-tool
$ sudo apt-get install -y jq python3-pip
$ sudo -H pip3 install yq
```

Link /bin/sh to bash

You must use bash instead of sh to build the code

```
$ sudo dpkg-reconfigure -plow dash
```

Choose '**No**' in the 'Configuring dash' popup dialog and press enter.

If it doesn't work, try:

```
$ sudo rm /bin/sh

$ sudo ln -s /bin/bash /bin/sh
```

- To ensure that the link was changed to bash:

```
$ ls -l /bin/sh
```

Build Source Code

ast26x0evb

1.1.3 ast2600evb

```
$ source setup evb-ast2600-insyde

$ bitbake obmc-phosphor-image
```

The output image with time stamp, "obmc-phosphor-image--xxxxxxxxxxxxxx.static.mtd, will be placed in:

/build/evb-ast2600-insyde/tmp/deploy/images/evb-ast2600-insyde

To find the latest image quickly, the latest output image is linked to the following synbolic link:

/build/evb-ast2600-insyde/tmp/deploy/images/evb-ast2600-insyde/image-bmc

1.1.4 ast2620evb

```
$ source setup evb-ast2620-insyde

$ bitbake obmc-phosphor-image
```

The output image with time stamp, "obmc-phosphor-image--xxxxxxxxxxxxxx.static.mtd, will be placed in:

/build/ evb-ast2620-insyde/tmp/deploy/images/evb-ast2620-insyde

To find the latest image quickly, the latest output image is linked to the following synbolic link:

/build/evb-ast2620-insyde/tmp/deploy/images/evb-ast2620-insyde/image-bmc

1.1.5 Compilation result

```
WARNING: obmc-phosphor-image-1.0-r0 do_rootfs: Group render has never been defined
WARNING: obmc-phosphor-image-1.0-r0 do_rootfs: Group sgx has never been defined
NOTE: Tasks Summary: Attempted 6952 tasks of which 12 didn't need to be rerun and all succeeded.

Summary: There were 66 WARNING messages.
benchang@BC /insyde_work/project/SDK/openbmc/build/evb-ast2600-insyde ((fad58ae33c...)) $ cat /etc/os-release
PRETTY_NAME="Ubuntu 22.04.5 LTS"
NAME="Ubuntu"
VERSION_ID="22.04"
VERSION="22.04.5 LTS (Jammy Jellyfish)"
VERSION_CODENAME=jammy
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=jammy
benchang@BC /insyde_work/project/SDK/openbmc/build/evb-ast2600-insyde ((fad58ae33c...)) $
```

Build BMC firmware for QEMU

- Since QEMU cannot simulate AST2600 perfectly, an user have to disable some unsupported hardware features to prevent QEMU from terminated abnormally.
- If an user would like to test firmware on QEMU environment, enable the flag IS_BUILDING_QEMU_IMAGE at local.conf file
- Take ast2600 evb as an example, after setup machine, enable the flag in /build/intel-ast2600-evb/conf/local.conf

```
# Enable this flag if an user wants to test firmware image on QEMU.
# With this flag some unsupported hardware feature will be disabled
# to prevent QEMU hang
IS_BUILDING_QEMU_IMAGE = "1"
```

Signing BMC Firmware

With Default Private Key

- OPF will use the following default private key to sign BMC firmware.
- The private key is used to sign the image.

<SDK>/meta-phosphor/recipes-phosphor/flash/files/OpenBMC.priv

- The publickey corresponding to private key is in the following location.
- The publickey is used to verify whether the image is valid or not when firmware update is procceeding.

<SDK>/build/tmp/work/$MACHINE-openbmc-linux-gnueabi/$RECIPE/$VERSION/rootfs/etc/activationdata/key/publickey

EX: (AST26x0)

<SDK>/build/tmp/work/evb_ast26x0_insyde-openbmc-linux-gnueabi/obmc-phosphor-image/1.0-r0/rootfs/etc/activationdata/OpenBMC/publickey

- If you decompress the BMC firmware image, there are some .sig files which are used to be verified with publickey

    1. image-full.sig
    2. image-kernel
    3. image-kernel.sig
    4. image-rofs
    5. image-rofs.sig
    6. image-runtime
    7. image-runtime.sig
    8. image-rwfs
    9. image-rwfs.sig
    10. image-u-boot
    11. image-u-boot.sig
    12. MANIFEST
    13. MANIFEST.sig

With Specific Private Key

- If you want to use other private key, not default one. You can use following way to produce your own private key.

:* Generate RSA1024 private key named "OpenBMC.priv"

```
# openssl genrsa -out OpenBMC.priv 1024
```

:* Generate RSA2048 private key named "OpenBMC.priv"

```
# openssl genrsa -out OpenBMC.priv 2048
```

:* Generate RSA4096 private key named "OpenBMC.priv"

```
# openssl genrsa -out OpenBMC.priv 4096
```

- Export some environment variables
- Build firmware image with environment variables

```
# export BB_ENV_EXTRAWHITE="$BB_ENV_EXTRAWHITE SIGNING_KEY"
# SIGNING_KEY=/path/to/secure/OpenBMC.priv bitbake obmc-phosphor-image (or intel-platforms)
```

- **It's a normal behavior when BMC Firmware update failed after private key has been changed.**

Build performance

- By default, Yocto use all CPU of the build machine to do the jobs.

If user want to specify the number of the jobs it can perform,

add following options to /conf/local.conf

```
BB_NUMBER_THREADS = "4"
PARALLEL_MAKE = "-j 4"
```

- Yocto uses build/downloads and build/sstate-cache as your default directories.

You can change them by adding the following options to build/conf/local.conf.

This will increase speed when you rebuild the same project with the same settings.

```
BB_GENERATE_MIRROR_TARBALLS = "1"
DL_DIR="/where/to/your/downloads"
SSTATE_DIR="/where/to/your/sstate-caches"
```

Troubleshooting

It is suggested that you double-check your versions of both xq and yq before you start to build OPF.
On Ubuntu 18.04:

```
$ xq --version
xq 2.12.0
$ yq --version
yq 2.12.0
```

On Ubuntu 20.04:

```
$ xq --version
xq 3.2.1
$ yq --version
yq 3.2.1
```

On Ubuntu 22.04:

```
$ xq --version
xq 3.2.2
$ yq --version
yq 3.2.2
```

Try upgrading your xq or yq if their versions are much lower than above. You may need to uninstall your local versions first.

```
$ pip3 uninstall xq
$ pip3 uninstall yq
$ sudo -H pip3 install yq
```

Then, exit and login again to take effect.